

NETWORKS AND COMPLEXITY

Solution 13-5

This is an example solution from the forthcoming book Networks and Complexity.

Find more exercises at <https://github.com/NC-Book/NCB>

Ex 13.5: Abstract two-dimensional map [Lvl. 2]

Consider the following system:

$$\begin{aligned}X_{i+1} &= -X_i + Y_i \\Y_{i+1} &= X_i - Y_i\end{aligned}$$

- a) Write the system in matrix form.

Solution

We write

$$\mathbf{x}_{i+1} = \mathbf{J}\mathbf{x}_i \quad (1)$$

where

$$\mathbf{J} = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \quad (2)$$

and $\mathbf{x}_i = (X_i, Y_i)^T$.

- b) Compute the eigenvalues of the matrix that appears in (a).

Solution

We write the characteristic polynomial

$$0 = (-\lambda - 1)(-\lambda - 1) - 1 \quad (3)$$

$$0 = (\lambda + 1)(\lambda + 1) - 1 \quad (4)$$

$$0 = \lambda^2 + 2\lambda \quad (5)$$

At this point we can see that $\lambda_1 = 0$ and $\lambda_2 = -2$.

- c) Find the corresponding eigenvectors and write the initial state $X_0 = 2$, $Y_0 = 0$ as a linear combination of eigenvectors.

Solution

For the first eigenvector we consider the first line of the matrix which leads to the condition

$$\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix} = 0 \begin{pmatrix} X \\ Y \end{pmatrix} \quad (6)$$

considering either line leads to

$$X = Y \quad (7)$$

and hence the eigenvector

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (8)$$

The second eigenvector must obey the condition

$$\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix} = -2 \begin{pmatrix} X \\ Y \end{pmatrix} \quad (9)$$

Considering the first line leads to

$$-X + Y = -2X \quad (10)$$

or in other words $Y = -X$ and hence to

$$\mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad (11)$$

or multiples thereof. With these eigenvectors it is straightforward to expand the initial state as

$$\begin{pmatrix} 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad (12)$$

(If you don't see this immediately, try the ansatz $\mathbf{x}_0 = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2$ then consider the lines separately. They will say $c_2 = c_1$ and $c_1 + c_2 = 2$ respectively which leads to $c_1 = c_2 = 1$.)

- d) Hence, solve the initial value problem (i.e. find X_i, Y_i for all $i > 0$).

Solution

For $i > 0$ we can write

$$\mathbf{x}_i = J^i \mathbf{x}_0 \quad (13)$$

$$= J^i (\mathbf{v}_1 + \mathbf{v}_2) \quad (14)$$

$$= J^i \mathbf{v}_1 + J^i \mathbf{v}_2 \quad (15)$$

$$= \lambda_1^i \mathbf{v}_1 + \lambda_2^i \mathbf{v}_2 \quad (16)$$

$$= (-2)^i \mathbf{v}_2 \quad (17)$$

$$(18)$$

Hence, the solution is

$$X_i = (-2)^i \quad Y_i = -(-2)^i \quad (19)$$

- e) Verify your solution by computing the first X_1, Y_1, X_2, Y_2 by hand.

Solution

Using the formula from the question

$$X_1 = -X_0 + Y_0 = -2 = (-2)^1 \quad (20)$$

$$Y_1 = X_0 - Y_0 = 2 = -(-2)^1 \quad (21)$$

$$X_2 = -X_1 + Y_1 = 2 + 2 = 4 = (-2)^2 \quad (22)$$

$$Y_2 = X_1 - Y_1 = -2 - 2 = -4 = -(-2)^2 \quad (23)$$

So, this works.

- f) Bonus: Using matrix methods, show that for any initial condition $X_i = -Y_i$ must hold for all $i \geq 1$.

Solution

For a given initial state $\mathbf{x}_0$ we can find c_1 and c_2 such that

$$\mathbf{x}_0 = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 \quad (24)$$

Starting from this state the general solution is

$$\mathbf{x}_i = \mathbf{J}^i \mathbf{x}_0 = c_1 \lambda_1^i \mathbf{v}_1 + c_2 \lambda_2^i \mathbf{v}_2 \quad (25)$$

Because $\lambda_1 = 0$, this simplifies to

$$\mathbf{x}_i = c_2 \lambda_2^i \mathbf{v}_2 \quad (26)$$

which means

$$X_i = c_2 (-2)^i \quad (27)$$

$$Y_i = -c_2 (-2)^i \quad (28)$$

And hence $X_i = -Y_i$.